

MEDI PAVAN KUMAR

SQL Engineer

+91-9392149379 | kumar2042000@gmail.com | Hyderabad, INDIA | Experience: 4 Years 9 Months | [LinkedIn](#)

PROFESSIONAL SUMMARY

Backend Infrastructure Engineer with 4+ years of experience building and operating production-grade backend systems, database platforms, automation pipelines, and cloud-native infrastructure. Specialized in Python backend engineering, Oracle/PostgreSQL performance tuning, Kubernetes-based deployments, and Linux platform operations.

Experienced in optimizing large-scale SQL workloads, automating operational systems, improving deployment reliability, and building internal platforms that reduce manual intervention and improve system resiliency. Strong foundation in backend services, CI/CD systems, observability, and infrastructure automation.

TECHNICAL SKILLS

Backend Engineering: Java, Python, FastAPI, Spring Boot, Flask, REST APIs, Backend Business Logic Implementation.

Databases & Performance Engineering: Oracle, PostgreSQL, MySQL, SQL, PL/SQL, Query Optimization, Indexing Strategies, Materialized Views, Execution Plan Analysis, Data Modeling

Cloud & Platform Engineering: AWS, ECS, EC2, Lambda, OpenShift, RDS, Apache ETL.

Observability & Operations: Prometheus, Cacti, Monitoring, Incident Reduction, System Reliability Engineering.

Security & Authentication: JWT, RBAC, Authentication & Authorization, Secure Deployment Practices.

Workflow & Automation: Apache Airflow, DAG Development, Python Automation Pipelines.

Tools & Collaboration: Git, Jira, Remedy, Agile/Scrum.

PROFESSIONAL EXPERIENCE (4+ YEARS)

SQL Engineer | iBasis India Pvt Ltd | Hyderabad | Apr 2023 - Present

Promoted to SQL Engineer from Solutions Engineer

Worked on enterprise-scale voice routing and pricing systems focused on backend workflows, database performance optimization, automation tooling, and scalable operational platforms.

Key Contributions:

- Built backend services and REST APIs using Java, Spring Boot, Python, Oracle, and PostgreSQL for enterprise pricing and routing systems.
- Optimized complex SQL and PL/SQL workloads, reducing execution times by up to 45% across reporting and operational workflows.
- Designed indexing, archival, and retention strategies supporting 3x data growth while maintaining platform stability and query performance.
- Automated operational workflows and internal tooling, eliminating 10-12 hours of weekly manual effort.
- Managed containerized workloads on Kubernetes and improved deployment reliability through CI/ CD enhancements using GitLab and Jenkins.
- Improved production observability using Prometheus and monitoring integrations, reducing MTTR by approximately 30%.
- Collaborated with cross-functional OSS and engineering teams to streamline integrations and accelerate issue resolution.
- Developed scalable database structures, including tables, views, and materialized views for pricing, routing, and customer deal management.
- Supported Linux-based production environments and backend operational troubleshooting.

Technologies Used: Python, Oracle, PostgreSQL, SQL, PL/SQL, Kubernetes, GitLab CI/CD, Jenkins, Linux, Airflow, Prometheus, Jira, Remedy

Build Engineer | PragmaEdge Software Services | Hyderabad | Sep 2021 - Apr 2023

Key Contributions:

- Standardized Jenkins and GitLab CI/CD pipelines, reducing build and deployment times by approximately 35%.
- Built automation workflows using Python, Ansible, and Terraform for repeatable infrastructure provisioning and application deployments.
- Implemented containerized deployment patterns across AWS, OpenShift, and Kubernetes environments.
- Automated rollback validation and deployment health checks, reducing production recovery windows from ~2.5 hours to ~28 minutes.
- Applied Linux hardening and secure deployment practices to improve system reliability and operational stability.
- Reduced production defects through improved deployment validation, automation, and release quality processes.
- Supported large-scale deployment automation for IBM and Sterling-based enterprise applications.

Technologies Used: Python, Linux, Jenkins, GitLab, Kubernetes, Ansible, Terraform, Oracle, AWS.

Web Developer Intern | Outshade | Hyderabad | Mar 2019 - May 2019

Key Contributions:

- Built **responsive web interfaces** and PHP-based components for dynamic applications, enhancing usability by 25%.
- Contributed to **code reviews and documentation**, reducing defects by 20% and improving team onboarding.

Technologies Used: PHP, MySQL.

PROJECTS

Voice Management Tool (VMT)

Enterprise Voice Routing & Pricing Platform | iBasis

Enterprise-scale Oracle-based platform for managing global voice traffic, customer pricing, floor costs, and bilateral routing operations.

Contributions:

- Designed and maintained Oracle tables, views, and materialized views supporting customer rates, sales pricing, floor pricing, and bilateral deal management.
- Developed backend services and REST APIs using Java, Spring Boot, Oracle, and PostgreSQL for pricing, routing, and operational workflows.
- Implemented PL/SQL procedures for pricing logic, markups, OBR processing, archival systems, and volume distribution workflows.
- Optimized complex SQL queries and execution plans, improving reporting and operational performance by approximately 45%.
- Built scalable backend business logic and database integration layers for enterprise voice management systems.
- Reduced storage utilization through archival and retention strategies across historical datasets.

Technologies: Java, Spring Boot, Oracle, PostgreSQL, SQL, PL/SQL

Customer Service Desk Automation

AI-Assisted Ticket Automation Platform | iBasis

Internal automation platform integrating Python workflows and LLM-assisted ticket triaging for customer and vendor issue management.

Contributions:

- Built Python-based automation workflows to process emails and operational requests for automated ticket creation
- Integrated LLM-assisted workflows to improve issue classification and response handling.
- Reduced manual operations by approximately 12–15 hours weekly and improved response times by ~35%.
- Improved workflow scalability and operational throughput through backend automation enhancements.

Technologies: Python, FastAPI, PostgreSQL, LLM Integration.

PSX Data Automation Suite

Deployment & Data Automation Platform | iBasis

Python-based internal platform for automating PSX deployments and operational workflows in Linux environments.

Contributions:

- Developed Flask-based backend APIs and workflow automation utilities.
- Improved query performance and reduced database load through optimization and indexing improvements.
- Automated multiple operational workflows, improving deployment visibility and reducing manual execution steps.

Technologies: Python, Flask, PostgreSQL, Linux

Build Automation

Infrastructure Automation System | PragmaEdge

Automation platform for deploying enterprise applications across AWS and OpenShift environments.

Contributions:

- Developed Terraform and Ansible automation pipelines for repeatable infrastructure provisioning.
- Automated installation, deployment, and configuration processes across AWS EC2, Lambda, ECS, and OpenShift.
- Reduced deployment time by approximately 70% and enabled faster release cycles.

Technologies: Python, Terraform, Ansible, AWS, OpenShift, ECS

UPS Logistics Platform Support

Secure Enterprise Integration & Deployment Operations | PragmaEdge

Supported deployment automation and secure enterprise data exchange systems using Sterling Integrator and Sterling File Gateway.

Contributions:

- Automated deployment and operational workflows for enterprise logistics systems.
- Configured and maintained secure communication protocols, including FTP, SFTP, AS2, HTTP, and HTTPS.
- Reduced partner onboarding time and improved operational reliability.

Technologies: Sterling Integrator, Linux, Automation Tools

EDUCATION

Bachelor of Technology in Computer Science Engineering | 07 2021

Sri Indu College of Engineering and Technology | Hyderabad

Class XII, PCM | 05 2017

Narayana Junior College | Hyderabad

Class X, SSC | 05 2015

Holy Mother High School | Hyderabad

CERTIFICATIONS

- Oracle APEX: Foundations
- Oracle SQL Performance Tuning, 2025
- Generative AI SQL Database Specialist with ChatGPT, 2025

ACHIEVEMENTS

- Best Employee of the Year at iBASIS, 2024
- Event Organizer in Space Apps India, 2018
- Social Media Volunteer in GES, 2017

LANGUAGES

- English
- Telugu
- Hindi